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Team Name:

Bin There, Done That

Project Name:

Smart parcel sorting system



SMART PARCEL SORTING SYSTEM

Date: June 12, 2025

LOGBOOK WEEK 7

1. Goal Past Week

- Start on the coding for state machine of STM32
- Draft wiring diagram of the system.
- Create a direct connection from Raspberry Pi 4 to STM32 and use AWSIoT as a log.

2. Achievement

- Drafted a new state machine for STM32 and Raspberry Pi 4 from Allan suggestion of connecting directly STM32 and Rasp 4 for camera usage.

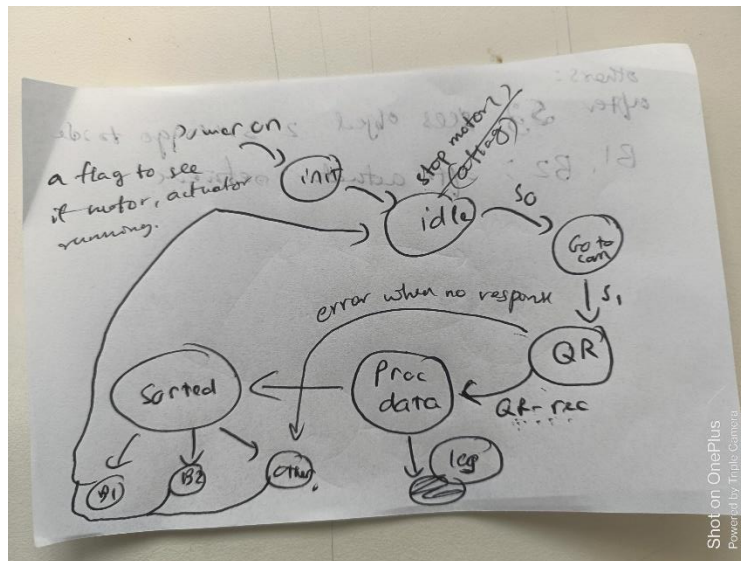


Figure 1: STM32's state machine

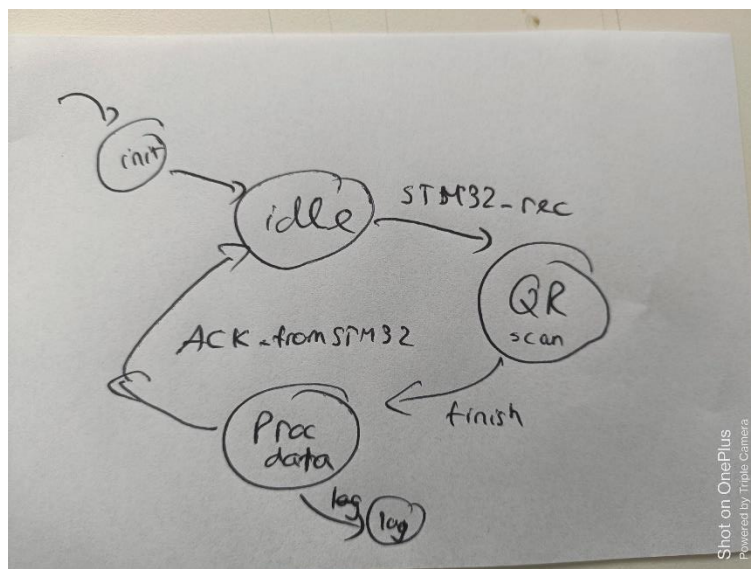
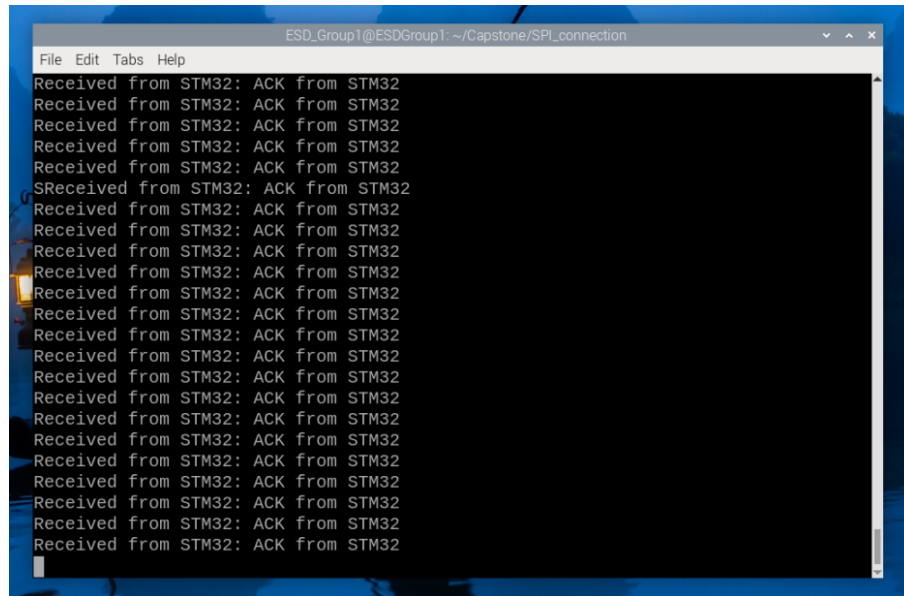


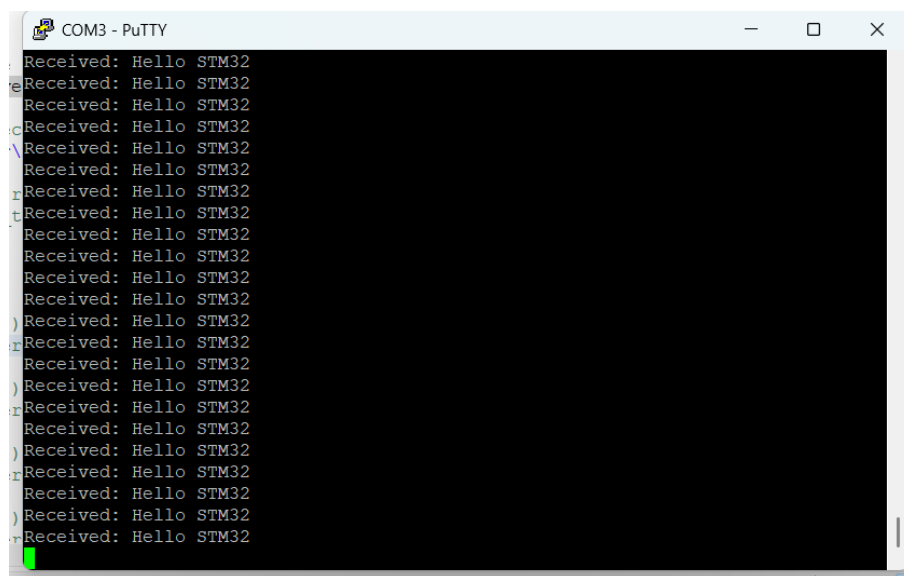
Figure 2: Raspberry Pi 4's state machine

- Initial software creation for SPI connection between STM32 and Raspberry Pi 4



```
ESD_Group1@ESDGroup1: ~/Capstone/SPI_connection
File Edit Tabs Help
Received from STM32: ACK from STM32
Received from STM32: ACK from STM32
Received from STM32: ACK from STM32
Received from STM32: ACK from STM32
Received from STM32: ACK from STM32
SReceived from STM32: ACK from STM32
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Received from STM32: ACK from STM32
Received from STM32: ACK from STM32
Received from STM32: ACK from STM32
Received from STM32: ACK from STM32
Received from STM32: ACK from STM32
```

Figure 3: Raspberry Pi received ACK from STM32



```
COM3 - PuTTY
Received: Hello STM32
Received: Hello STM32
Received: Hello STM32
Received: Hello STM32
Received: Hello STM32
Received: Hello STM32
Received: Hello STM32
Received: Hello STM32
Received: Hello STM32
Received: Hello STM32
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Received: Hello STM32
Received: Hello STM32
Received: Hello STM32
Received: Hello STM32
Received: Hello STM32
```

Figure 4: STM32 received message from Raspberry Pi 4

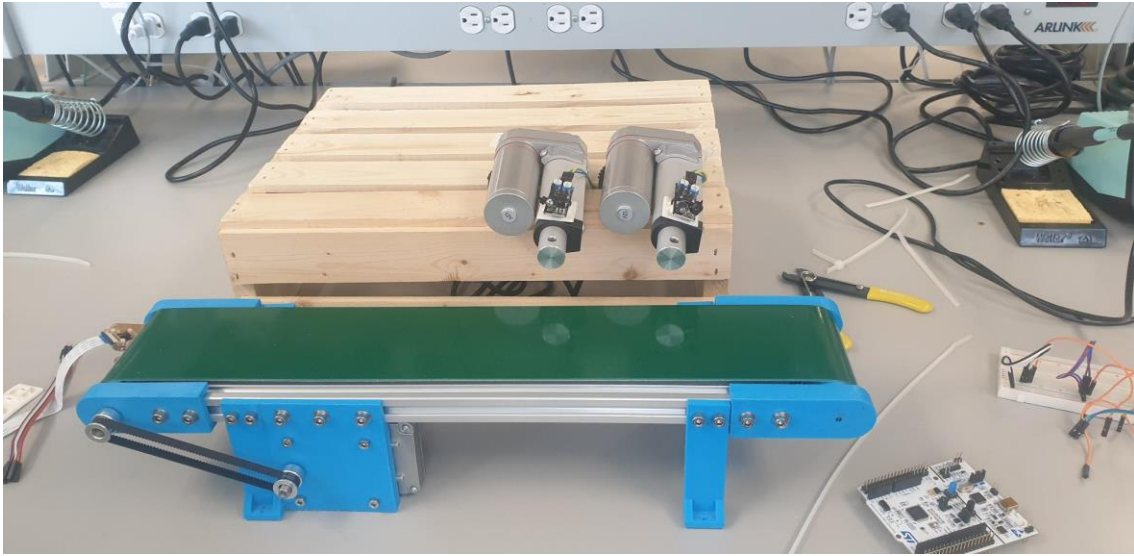


Figure 5: Mounted hardware (In progress)

2.1. Problems occur:

No problem

2.2. Resolve:

Not Applicable

3. Goal for Week 8

- Continued to mount all the hardware.
- Start coding for the new state machine
- If time permits, start coding and testing on all of the mounted hardware.